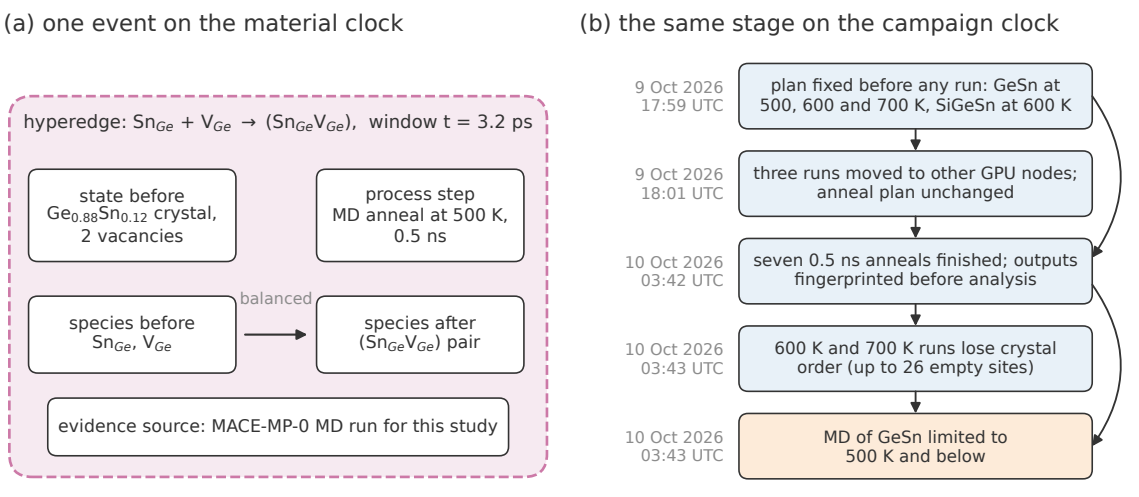
(a) one event on the material clock
hyperedge: Sn_{Ge} + V_{Ge} → ($Sn_{Ge}V_{Ge}$), window t = 3.2 ps
state before $Ge_{0.88}Sn_{0.12}$ crystal, 2 vacancies
process step MD anneal at 500 K, 0.5 ns
species before Sn_{Ge}, V_{Ge}
balanced
species after ($Sn_{Ge}V_{Ge}$) pair
evidence source: MACE-MP-0 MD run for this study
(b) the same stage on the campaign clock
9 Oct 2026 17:59 UTC
plan fixed before any run: GeSn at 500, 600 and 700 K, SiGeSn at 600 K
9 Oct 2026 18:01 UTC
three runs moved to other GPU nodes; anneal plan unchanged
10 Oct 2026 03:42 UTC
seven 0.5 ns anneals finished; outputs fingerprinted before analysis
10 Oct 2026 03:43 UTC
600 K and 700 K runs lose crystal order (up to 26 empty sites)
10 Oct 2026 03:43 UTC
MD of GeSn limited to 500 K and below